

Detailed Workshop Schedule

From Cells to Circuits | 5-Day Biomedical Engineering Workshop | July 7–11, 2026

 Workshop: **7–11 July 2026**

 Daily: **10:00 AM – 1:00 PM**

 Venue: **Shobhit University Campus, Meerut**

 **Class XII Science Students**

📌 **Important Note:** All sessions are held daily from **10:00 AM to 1:00 PM**. Participants are requested to arrive by 9:50 AM. Lab coats and ID cards must be worn in all laboratory sessions. All materials, lab consumables, and refreshments are provided. Please carry a notebook and pen each day.

WORKSHOP DURATION
7 – 11 July 2026 (5 Days)

DAILY TIMINGS
10:00 AM – 1:00 PM

VENUE
BME Labs, Block — 5, Shobhit University

ELIGIBILITY
Class XII Science Students (PCB / PCM)

Day 01

The Living Blueprint
Cell Biology & Microscopy

 Tue, 7 Jul 2026

TIME	ACTIVITY	DETAILS	TYPE
10:00 – 10:30 AM	Welcome Address & Workshop Overview	Orientation, introduction to faculty & safety briefing	Activity
10:30 – 11:00 AM	Lecture: Introduction to Biomedical Engineering	History, scope & career pathways in BME	Lecture
11:00 – 11:50 AM	Lab: Microscopy & Cell Observation	Prepare slides of human cheek cells; observe under compound microscope; identify organelles	Lab
11:50 AM – 12:00 PM	Refreshment Break	Tea & light snacks	Break
12:00 – 12:45 PM	Lab: Blood Grouping (ABO System)	Use antisera to identify ABO blood group; understand agglutination reaction	Lab
12:45 – 1:00 PM	Day Debrief & Q&A	Reflection, doubt clearing, distribute Day 2 preparation notes	Activity

 **Day Outcome:** Students will have prepared microscopy slides, identified blood groups, and understood the fundamental unit of life — the cell.

Day 02

Signals of Life

Biosignals — ECG, EMG & Sensors

Wed, 8 Jul 2026

TIME	ACTIVITY	DETAILS	TYPE
10:00 – 10:20 AM	Quick Recap & Concept Check	5-minute quiz on Day 1 content; discuss observations	Activity
10:20 – 11:00 AM	Lecture: Bioelectrical Signals in the Human Body	ECG, EEG, EMG fundamentals; how the heart generates electrical signals	Lecture
11:00 – 11:50 AM	Lab: ECG Recording & Analysis	Attach electrodes; record 12-lead ECG on volunteers; measure heart rate; identify P, QRS, T waves	Lab
11:50 AM – 12:00 PM	Refreshment Break	Tea & light snacks	Break
12:00 – 12:45 PM	Demo: Biosensor Technology	SpO ₂ sensor, temperature sensor, skin conductance demo; understand transducer principles	Demo
12:45 – 1:00 PM	Day Debrief & Q&A	Journal entry; group discussion on clinical significance of ECG	Activity

 **Day Outcome:** Students record real ECG traces, identify cardiac waveforms, and understand how biological signals are captured by sensors.

Day 03

From Signal to Circuit

Electronics & Amplifier Design

Thu, 9 Jul 2026

TIME	ACTIVITY	DETAILS	TYPE
10:00 – 10:20 AM	Recap: Biosignals Review	Peer teaching — students explain ECG concepts to each other	Activity
10:20 – 11:00 AM	Lecture: Biomedical Circuit Design	Op-amp basics, instrumentation amplifier, noise filtering in medical devices	Lecture
11:00 – 11:50 AM	Lab: Build an Instrumentation Amplifier	Breadboard wiring; connect INA128 chip; test with low-voltage signal; measure gain on oscilloscope	Lab
11:50 AM – 12:00 PM	Refreshment Break	Tea & light snacks	Break
12:00 – 12:45 PM	Lab: Connect Sensor to Amplifier	Interface ECG electrodes with amplifier circuit; observe signal on oscilloscope; adjust gain	Lab
12:45 – 1:00 PM	Day Debrief & Q&A	Circuit troubleshooting discussion; journal entry on today's build	Activity

 **Day Outcome:** Students wire a functional instrumentation amplifier and connect it to biological sensors — bridging biology and electronics.

TIME	ACTIVITY	DETAILS	TYPE
10:00 – 10:20 AM	Recap: Circuit Review	Live debugging session; students fix each other's circuits	Activity
10:20 – 11:00 AM	Lecture: Embedded Systems for Healthcare	Arduino platform, IoT in medicine, real-time data acquisition	Lecture
11:00 – 11:50 AM	Lab: Arduino Programming	Write sketch to read analog sensor; display on serial monitor; control LED with biosignal threshold	Lab
11:50 AM – 12:00 PM	Refreshment Break	Tea & light snacks	Break
12:00 – 12:45 PM	Lab: MATLAB Signal Visualization	Import ECG data; plot waveforms; apply digital filter; compute heart rate variability metrics	Lab
12:45 – 1:00 PM	Day Debrief & Project Briefing	Introduce Day 5 Innovation Challenge; form teams	Activity
Day Outcome: Students program an Arduino to acquire sensor data and use MATLAB to visualize and analyze biological signals.			

TIME	ACTIVITY	DETAILS	TYPE
10:00 – 10:30 AM	AI in Healthcare — Demo & Discussion	AI-based ECG diagnosis demo; ML model for disease prediction; discuss ethical AI in medicine	Demo
10:30 – 11:45 AM	Innovation Challenge Presentations	Teams present a 5-minute solution to a real-world biomedical problem using workshop concepts	Activity
11:45 AM – 12:00 PM	Refreshment Break	Tea & light snacks	Break
12:00 – 12:30 PM	Faculty Feedback & Career Guidance	B.Tech BME programs, research opportunities, GATE, international pathways	Lecture
12:30 – 1:00 PM	Certificate Distribution Ceremony	Certificates awarded to all participants by faculty; group photograph; feedback forms	Ceremony
Day Outcome: Students showcase their innovation, receive certificates, and leave with a clear understanding of careers in biomedical engineering.			

- What to Bring Each Day:** College/school ID card · Lab coat (available at campus if needed) · Notebook & pen · Water bottle · Comfortable footwear
- What Is Provided:** Lab consumables · Reference materials · Refreshments · Lab coat (if needed)